EU SAE WORKBENCH · GUIDE 3 OF 3

Estimating poverty, anchored poverty and mean welfare

Three worked examples: what to set, and how to read the results.



OVERVIEW

Three examples, one workflow

Each example follows the seven steps of Guide 2. Only Steps 3 and 4 change.

EXAMPLE 1

Poverty headcount

The share of people below each year's poverty line, for example the at-risk-of-poverty threshold.

EXAMPLE 2

Anchored poverty

A poverty line held fixed in real terms, so that the standard does not move with incomes.

EXAMPLE 3

Mean welfare

Average income or consumption per person, in the prices of one year.

Data: Data/simulated (40 domains, survey years 2022 and 2023; incomes treated as those of the year before). The same steps apply to real survey data.

Years and prices: five terms

Files are keyed by survey year. Price indices and poverty lines depend on the year the incomes refer to.

Term	Meaning	Example
Survey year	The year chosen in Step 3. Survey, auxiliary, population and target files use it	2022, 2023
Income reference year	The year the incomes refer to. In EU-SILC usually the year before the survey	2021, 2022
CPI reference year	The year in which the index equals 100. It does not change the results	2015 = 100
Price base year	The year whose prices real welfare is reported in	2021 prices
Anchor year	The year whose poverty line is held fixed in real terms	2021 line: 10,000

Only ratios between years matter. CPI 2021 = 110.0 and CPI 2022 = 114.4: welfare of 10,920 in 2022 is 10,920 × 110.0 / 114.4 = 10,500 in 2021 prices.

EXAMPLE 1 OF 3

Poverty headcount with year-specific poverty lines

Each survey year has its own line, for example 60% of that year's median income.

EXAMPLE 1 · CONCEPT

What the poverty headcount measures

The population-weighted share of people whose welfare is below the poverty line of their year.

The measure

FGT(0), the headcount ratio. FGT(1) and FGT(2) measure the depth and severity of poverty.

The poverty line

One line per survey year, in the same currency, period and welfare units (per person or equivalised) as the survey.

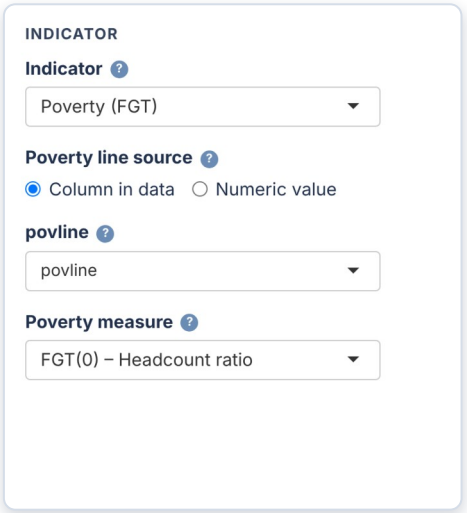
EU at-risk-of-poverty

60% of the national median equivalised disposable income. In EU-SILC it refers to the previous year's incomes.

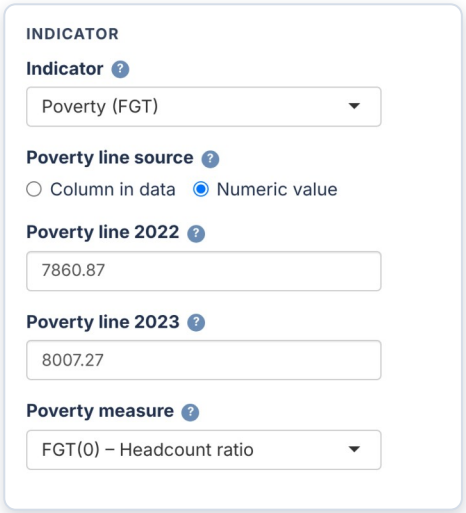
The Workbench uses the lines you supply; it does not calculate national medians. A change in the rate reflects changes in both incomes and the line.

Two ways to give the poverty lines

With the same lines, both give the same estimates. Choose Poverty (FGT) and FGT(0) – Headcount ratio.



Column in data: one line per record



Numeric value: one box per survey year

Simulated data: the povline column is 60% of each year's person-weighted median income: 7,860.87 in 2022 and 8,007.27 in 2023.

Model settings for poverty rates

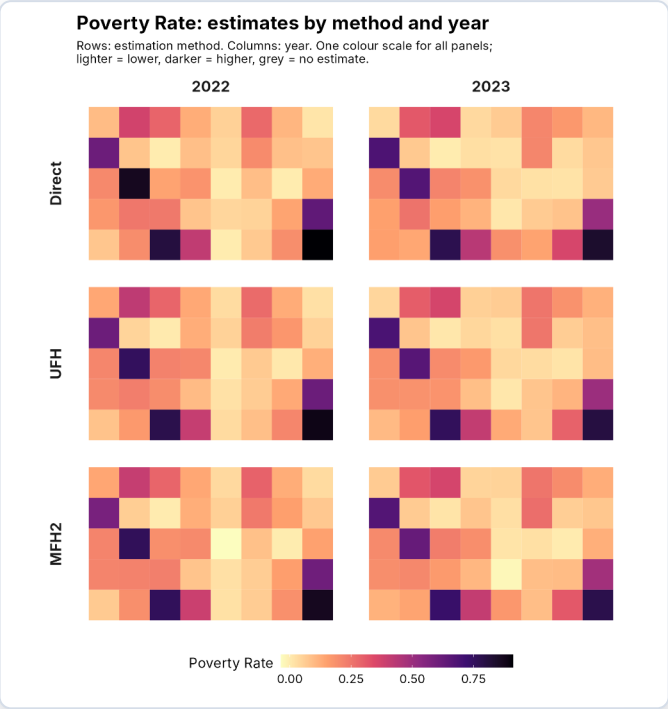
New runs use the defaults below. Saved choices are retained; examples remain historical.

Setting	Default	Why
UFH transformation	arcsin, with bias correction (bc)	Keeps rates between 0 and 1 and stabilises the variances. For FGT(1) and FGT(2) the app uses 'no'.
UFH model selection	BIC	Prefers simpler models
MFH model	MFH2	Both years jointly, with correlated area effects
MFH variances and covariance	sm_out, rho_sm_out	Direct variances (GVF below 0.001) with national sampling correlation
Bootstrap replicates	500	Uncertainty of levels and changes
Benchmarking	off	If on, targets must use the same poverty lines

If an MFH variance or correlation ends at its limit, the report flags it and change significance is provisional.

Reading the results

Historical w5o: sm_out / rho_sm_out (current defaults), 100 replicates.



22.0% → 21.0%

National direct headcount, 2022 to 2023: −1.0 percentage point

Rows: Direct (survey only), UFH and MFH2. Model estimates are more stable where samples are small.

Changes: use percentage points. Counts of pointwise-significant areas are exploratory. BH and Bonferroni results are also available.

Files: estimates with RMSE and CV in outputs/data/pov_comparison_detailed.xlsx.

EXAMPLE 2 OF 3

Anchored poverty

Keep the poverty standard of one year fixed in real terms, and supply it for every survey year.

EXAMPLE 2 · CONCEPT

A poverty line fixed in real terms

The line keeps the purchasing power of the anchor year's threshold.

Why anchor

With a relative line, poverty can stay flat while real incomes fall. A fixed real standard shows absolute changes in living standards.

Where the lines come from

Use official anchored thresholds where available. Check the anchor year, the income years, the currency and the welfare units.

Two ways to enter them

Numeric value: one line per survey year in Step 3. Column in data: a survey column, when lines vary across records.

Do not recalculate the line from each later year's median: that would move the standard.

Anchored lines for each survey year

Illustration: a 2021 line of 10,000, kept fixed in real terms with the price index of Example 3.

Survey year	Income year	Price index	Anchored line
2022	2021	100.00	10,000
2023	2022	104.00	10,400
2024	2023	106.08	10,608

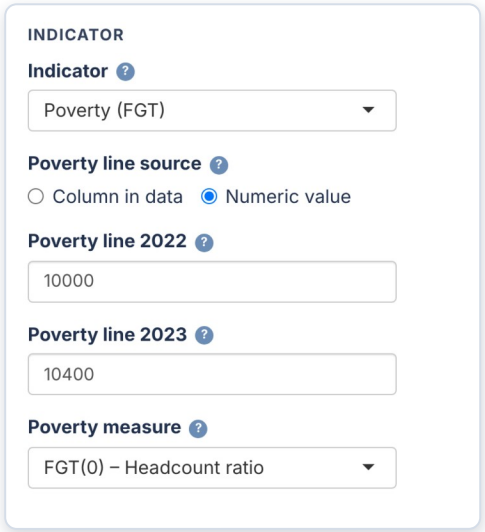
Line = 10,000 × index(income year) / index(2021)
Survey 2023 asks about 2022 incomes, so it gets the 2022 line.

If the survey's welfare is already in 2021 prices, the anchored line is 10,000 in every year.

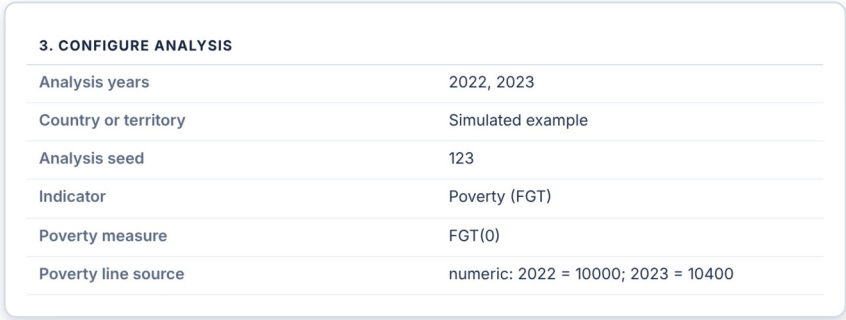
These lines are illustrative. They are not the at-risk-of-poverty lines of the simulated data.

Enter the anchored lines and check them

Poverty (FGT), FGT(0), Numeric value. No price conversion: the lines are already in each year's prices.



Step 3: the anchored lines typed in



Step 6: the summary repeats them

Then check data readiness and run the model as in Example 1. Report anchored rates and their changes in percentage points.

EXAMPLE 3 OF 3

Mean welfare in real terms

Average income or consumption per person, in the prices of one year.

Two routes to real mean welfare

Changes in current-price welfare include inflation. Choose how prices are handled in Step 3.

Route A: welfare already in constant prices

- The survey's welfare was deflated before you received it.
- Step 3: constant (real) prices already. No price index is needed.
- The year of those prices is optional and used in labels only.

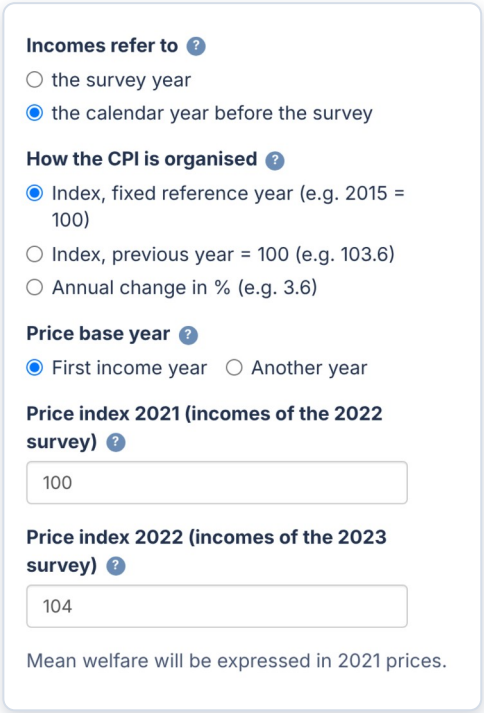
Route B: current prices plus a price index

- Welfare is in each year's own prices.
- Step 3: current prices: convert with a price index.
- Type the index by year, or read it from the CPI file of Step 1.

The default, current prices without conversion, reports nominal changes. It is not one of the two real-welfare routes.

Converting with a price index

Simulated example: incomes of the year before the survey, expressed in 2021 prices.



Setting	Choice
Indicator	Mean welfare
Welfare in the survey data is in	current prices: convert with a price index
Price index values	Type them in by year
Incomes refer to	the calendar year before the survey
How the CPI is organised	Index, fixed reference year
Price base year	First income year (2021)
Price index 2021, 2022	100 and 104

The app confirms: “Mean welfare will be expressed in 2021 prices.”

Three accepted CPI formats

Use one national annual-average series. The three rows below describe the same inflation.

How the CPI is organised	2021	2022	2023
Index, fixed reference year	100.00	104.00	106.08
Index, previous year = 100	—	104.0	102.0
Annual change in %	—	4.0	2.0

Fixed index: give the income years and the price base year.

Annual formats: give every annual change up to the latest year needed; the app chains them.

CPI file: one row per year with a year column and the index (decimal commas are accepted).

```
year,cpi
2021,100.00
2022,104.00
```

An illustration in 2021 prices

Each record is converted before the area means are calculated.

Survey / income year	Price index	Mean in current prices	Mean in 2021 prices
2022 / 2021	100.00	10,400	10,400
2023 / 2022	104.00	10,920	10,500

Real welfare = current-price welfare × index(2021) / index(income year)

+5.00%
change in current prices

+0.96%
real change

Illustrative numbers, not results of the simulated data.

Model settings for mean welfare

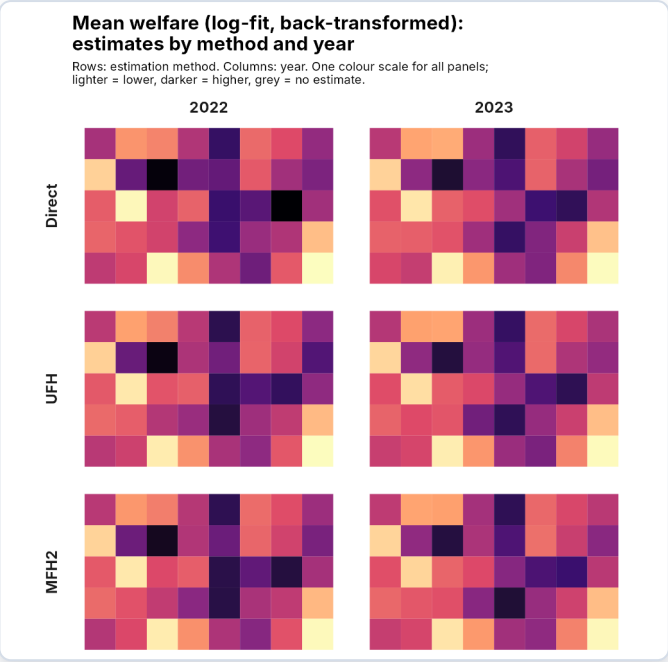
Log is the default for mean welfare; no transformation remains available. Deflate first.

Setting	Choice	What it means
UFH transformation	log	Models the log of each area's direct arithmetic mean
UFH bias correction	bc_sm	Slud–Maiti correction (emdi) when returning to currency units
MFH transformation	log	The same, for both years jointly
MFH bias correction	Conditional lognormal	exp(η + v/2), v the conditional random-effect variance
Working variances	sm_out	UFH log honors the selector; MFH uses rho_sm_out

Zero and negative incomes stay in the direct means. A log model needs every area mean to be positive; the readiness check (Test 0e) reports this.

Reading the results

Historical w5o, Route B: sm_out / rho_sm_out (current defaults), 100 replicates.



Lighter areas: lower mean welfare; darker: higher (2021 prices).

15,079 → 14,674

National direct mean welfare in 2021 prices: −2.7% real change

Output	Unit
Levels and RMSE	Currency, in 2021 prices
Changes and their tests	Percentage change of the final means
Change-and-RMSE box plots	ln(mean 2023 / mean 2022), not × 100
Excel columns ending in _eur	Currency units, whatever the currency

Checks before you report any of the three

The model can only be as consistent as its inputs.

1. **Same concept**
   Welfare, poverty lines and benchmark targets use the same definition, units and prices.

2. **Readiness**
   No ERROR; warnings read and understood, including the weighted population total (Test 0f).

3. **Diagnostics**
   Model fit, bootstrap success and Monte Carlo precision in the report.

4. **Boundaries**
   If an MFH variance or correlation is at its limit, change significance is provisional.

5. **Bootstrap**
   Use the default 500 replicates or more for results you report.

6. **Interpretation**
   Interval coverage is not certified. Compare Direct, UFH and MFH before concluding.

Revise each annual estimate once

T = 2 supports this NSO policy. Freezing estimates is not automatic.

Window	Previously reported year	New year
2022, 2024	Decide the initial 2022 status	2024 provisional
2024, 2025	Revise 2024 once, then freeze	2025 provisional
2025, 2026	Revise 2025 once, then freeze	2026 provisional

Revision burden: at most one prior year per release. Longer windows can repeatedly revise several overlapping years.

NSO responsibility: enforce freezes, document exceptional reopening, and archive each window and publication vintage.

Change uncertainty: retain each pair's estimates and MSE/MCPE together. Within-pair uncertainty does not cover differences across frozen vintages.

MORE DETAIL

Where to read further

Guidelines note: methods, years and prices, and reading the change outputs.
Methodology mapping and validation status: in Guides and notes, inside the Workbench.

Questions and problem reports: github.com/Noboyoshida227/eu-sae-application/issues